MILITARY SPECIFICATION

PUMP, VACUUM, POSITIVE DISPLACEMENT, AIR ROTARY
VANE, TYPE 2 CAST ALUMINUM

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the pumps described herein shall consist of this document and the latest issue of Specifications MIL-P-28824.

1. SCOPE.

1.1 Scope. This specification covers the detail requirements for type 2, rotary vane air pump with cast aluminum housing.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the Following specifications and standards of the issue listed in that issue of the Department Of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specifications to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-8625		Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024/5	-	Plate, Identification

STANDARDS

MILITARY

MS33537	Helical Coil Installation Thread, Design Standard For

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander Naval Electronic Systems Command (ELEX 8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 4310

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with specification MIL-P-28824 and as specified herein.

3.2 Materials. The materials shall be as specified in Figures 2 through 8.

3.3 Details of components. The components shall be as specified in Figures 1 through 8.

3.4 Design and construction. The design and construction shall be as specified in Figures 1 through 8.

3.5 Dimensions. The dimensions shall be in accordance with Figures 3 through 8. The dimensions shall not vary outside the tolerances given within the normal mean-time-between-failure (MTBF), except that surfaces in contact with carbon vanes may show expected wear characteristics only.

3.5.1 Alignment. Rotor alignment shall be checked by removing the end cap retainers and shims and checking the end play. The end play reading shall be at least .003 inches. The rotor shall be centered between the end plates by shiming the bearings.

3.5.2 Tolerances. Rotor and body tolerances shall be such that the rotor shall not contact the body or end plates when operated at 65°C.

3.6 Color and finish.

3.6.1 External. The pump shall be hard anodized in accordance with MIL-A-8625 Type III to .004 inch thickness.

3.6.2 Internal. Internal surfaces in contact with moving vanes shall be hard anodized in conformance with MIL-A-8625 Type III. Hard anodizing shall be .004 inch thick and finished in accordance with Figures 3, 4, and 5.

3.7 Marking. Marking shall be in accordance with MIL-P-15024/5. One decal may be affixed to indicate tampering and if used it shall contain a warranty note when warranty or other guarantee is stipulated in the contract. The warranty note shall be covered with clear lacquer and shall state, “This pump is under warranty until (month, year)“. It shall not obstruct other required markings as shown in Figure 1.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Sampling and inspection shall be in accordance with specification MIL-P-28824 and as specified herein.

4.2 Teardown inspection. The teardown inspection shall, as a minimum, conform to the requirements of Appendix A. A teardown inspection shall be made as required and at the end of every reliability test. All dimensions and surfaces shall be checked to ensure that they are within the tolerances given in Figures 3 through 8.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation-packaging. Preparation for delivery shall be in accordance with specification MIL-P-28824.

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6. NOTES

6.1 The notes specified in specification MIL-P-28824 are also applicable to this specification.

6.2 Test equipment. The test equipment recommended for use in the tests given in Appendix A is listed in Table I of Appendix A.

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APPENDIX A

INPSECTION PROCEDURES, REQUIREMENTS, AND TEST EQUIPMENT

10. SCOPE

10.1 Scope. This appendix details the inspection procedures for end housings, pump body, and pump rotor. It also includes inspection requirements and equipment.

20. INSPECTION PROCEDURE FOR END HOUSING.

20.1 Position air leaf head in seal bore of end housing and record the indicator reading.

20.2 Posit Ion air leaf head in bearing bore of end housing and record the indicator reading.

20.3 Remove air leaf head from bearing bore.

20.4 Measure the depth of bearing and seal shoulder from outside surface with a depth and record readings.

20.5 Measure and record the shoulder width.

20.6 Measure and record the end housing width and height.

20.7 Measure and record the diameter of the bearing end of the end housing.

20.8 Measure and record the bolt circle diameter for the mounting holes and bearing retainer.

20.9 Measure and record the dowel pin radius.

20.10 Place end housing on a surface plate with the bearing end down and position air test indicator (ATI) gage head stylus on Surface A. Move stylus about surface and record ATI reading for flatness of Surface A.

20.11 Position end housing on fixture with Surface A down.

20.12 Position the ATI stylus on the top surface of bearing bore. Rotate end plate 360° and record total indicator reading (TIR) .

20.13 Position the ATI head stylus on the inside top edge of bearing bore on side nearest ATI.

20.14 Rotate the end of plate 360°. Record the difference of maximum and minimum readings.

20.15 Adjust ATI stylus downward to the bottom of the bore, just above the shoulder.

20.16 Repeat Step 20.13.

20.17 Reverse end housing on fixture.

20.18 Repeat Steps 20.12 through 20.15 for seal bore.

20.19 Position the ATI stylus on bearing shoulder. Rotate end plate 360° and record total indicator reading.

20.20 Measure finish with surface analyzer.

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20.21 Check hardcoat for continuity and thickness.

20.22 Check Helicoil installation in accordance with MS 33537.

30. INSPECTION PROCEDURES FOR PUMP BODY

30.1 Set up body bore plunge jet head to masters (high and low limits) per instruction manual. (This shall be performed once daily prior to test).

30.2 Set ATI to standard body width with the block on base surface of plunge jet head. Readjust ATI column calibration as required.

30.3 Move ATI back from block and remove gage block.

30.4 Place pump body on plunge jet head.

30.5 Place ATI stylus on face of pump body. Rotate body 360° and record maximum and minimum ATI reading for body width. The difference between these maximum and minimum readings is the parallelism of the two faces.

30.6 Remove ATI from contact with pump body.

30.7 Record Column 1 reading for body bore diameter.

30.8 Rotate body until the float in Column 2 reaches its highest position. Align upper black line on sliding indicator with the top of float.

30.9 Rotate body 360°. The float shall remain within the indicated tolerance for squareness.

30.10 Reverse body on fixture and repeat Steps 30.8 and 30.9.

30.11 Measure and record the body O ring groove inside and outside diameters and depth on both the drive end and antidrive end.

30.12 Measure and record the foot length and width and the foot mounting hole length and width.

30.13 Measure and record the diameters of the foot mounting holes.

30.14 Measure and record the body height and overall hex.

30.15 Measure and record the body radius and bore height to center line.

30.16 Measure and record the diameter and threads of the inlet and outlet ports.

30.17 with the body positioned on a surface plate with Surface A up, position ATI stylus on Surface A.

30.18 Rotate the body 360° and record the flatness of Surface A.

30.19 Measure finish with surface analyzer.

30.20 Check hardcoat for continuity and thickness.

30.21 Check Helicoil installation in accordance with MS 33537.

40. INSPECTION PROCEDURE FOR PUMP ROTOR

40.1 Set up equipment to masters (high and low limits) per instruction manual. (This shall be performed once daily prior to test).

40.2 Measure rotor diameter with the “C” snap air and record reading

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of Column 5.

40.3 Pleasure rotor width with the “C” snap air and record reading of Column 6.

40.4 Measure rotor bearing Journal diameters (drive and antidrive) with the "C" snap air and record reading of Column 7.

40.5 Measure rotor seal seat diameters (drive and antidrive) with the “C” snap air and record readings of Column 8.

40.6 Measure and record the total rotor shaft length and the short shaft length.

40.7 Measure and record the shaft diameter.

40.8 Measure and record the snap ring groove diameter and width.

40.9 Measure and record seal journal length and bearing journal length.

40.10 Measure and record the overall length from bearing journal to bearing Journal.

40.11 Measure and record the length and depth of the flat on the drive end of the rotor shaft.

40.12 Measure and record the distances on the shaft between the center drill indentations and the bearing journals.

40.13 Measure and record the slot width, depth and distance from center line.

40.14 Position rotor in V block on bearing journals.

40.15 Place ATI stylus on diameter of rotor at extreme drive end of rotor. Rotate the rotor 360° and record the TIR for rotor runout.

40.16 Repeat Step 40.15 for the antidrive end.

40.17 Place ATI stylus on seal Journal of the drive end of the rotor. Rotate the rotor 360° and record the TIR for the seal journal runout.

40.18 Repeat Step 40.17 for the antidrive end.

40.19 Place stylus on shaft on drive end of rotor. Rotate the rotor 360° and record the TIR for shaft runout.

40.20 Repeat Step 40.19 for the antidrive end.

40.21 Check markings, nameplates, and Warning decals on pump as per drawing.

50. INSPECTION REQUIREMENTS

50.1 Inspection equipment. The equipments utilized in this procedure and listed by make or model number carry no implication of preference, recommendation or approval by the Department of Defense for use. It is recognized that equivalent equipment produced by other manufacturers may be capable of equally satisfactory performance in this procedure. The inspection equipment and fixtures required for inspection of the vacuum pump are listed in Table I.

50.2 Inspection conditions. Inspections are to be made in a controlled environment at local atmospheric pressure and controlled temperature of 23°C±1. Relative humidity will be between 40 and 65 percent during measurements.

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TABLE I. Inspection equipment

Manufacturer and Model Number	Description	Minimum Use Specifications
Bendix Model M08SM	Precisionaire 8 column instrument	Column 1 5000:1 “O” center scale 2 20,000:1 No scale 3 & 4 5000:1 “O” center scale 5 & 6 2000:1 “O” center scale 7 & 8 5000:1 “O” bottom scale
Bendix Model M01-5M	Precisionaire 1 column instrument	5000:1 “O” center scale
Bendix No. 60053834	Airtest indicator kit	None
Bendix No. 60053827	Set up block master	None
Bendix No. 60108081	Gage stand	None
Bendix No. 60055402		#1 2.1185-2.1195 width #2 2.8065-2.8075 O.D.
Bendix No. 60087000	Fixed	#l 0.5905-0.5907 O.D. #2 0.7491-0.7511 O.D.
Bendix No. 2472290-13	Blind type leaf jet spindle	1.3745-1.3755 I.D.
Bendix No. 24722290-14	Blind type leaf jet spindle	1.3745-1.3785 I.D.
Bendix No. 2472290-2	Plunjet gaging spindle	#l 3.2465-3.2565 I.D. #2 90 ° 1 minute squareness of bore to face
Bendix No. 2472290-20	Double diameter truax arbor assembly	As specified
Bendix AGD, Style #3 class X	Master setting disc	Min. size 0.5905 dia. Max. size 0.5907 dia.
Bendix AGD, Style #3 Class X	Master setting disc	Min. size 0.7491 dia. Max. size 0.7511 dia.

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TABLE I. Inspection equipment

Manufacturer and Model Number	Description	Minimum Use Specifications
Bendix AGD, Style #3 Class X	Plaster setting disc	Min. size 2.1185 dia. Max. size 2.1195 dia.
Bendix AGD, Style #3 class X	Master setting disc	Min. size 2.8065 dia. Max. size 2.8075 dia.
Bendix AGD, Class X	Master ring gage	Min. size 1.3745 dia. Max. size 1.3755 dia.
Bendix AGD, Class X	Master ring gage	Min. size 1.3775 dia. Max. size 1.3785 dia.
Bendix No. 24722290-3	Set of masters	Set-up masters for No. 2472290-2
Vernier Calipers		0-6
Depth Micrometers		0-1
Micrometers		0-1
Micrometers		1-2
Gage Blocks	Set	Do all Model 86-S

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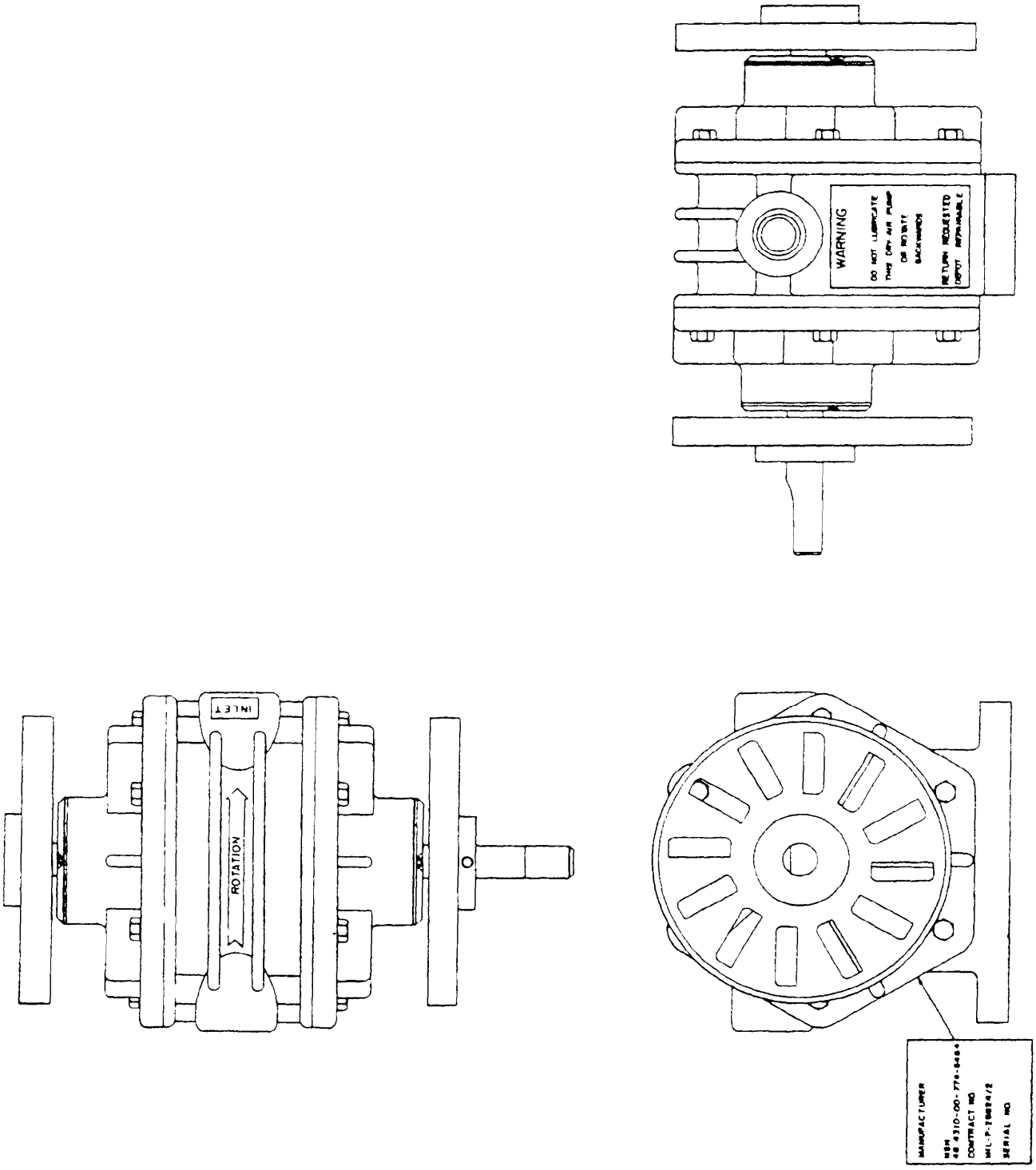
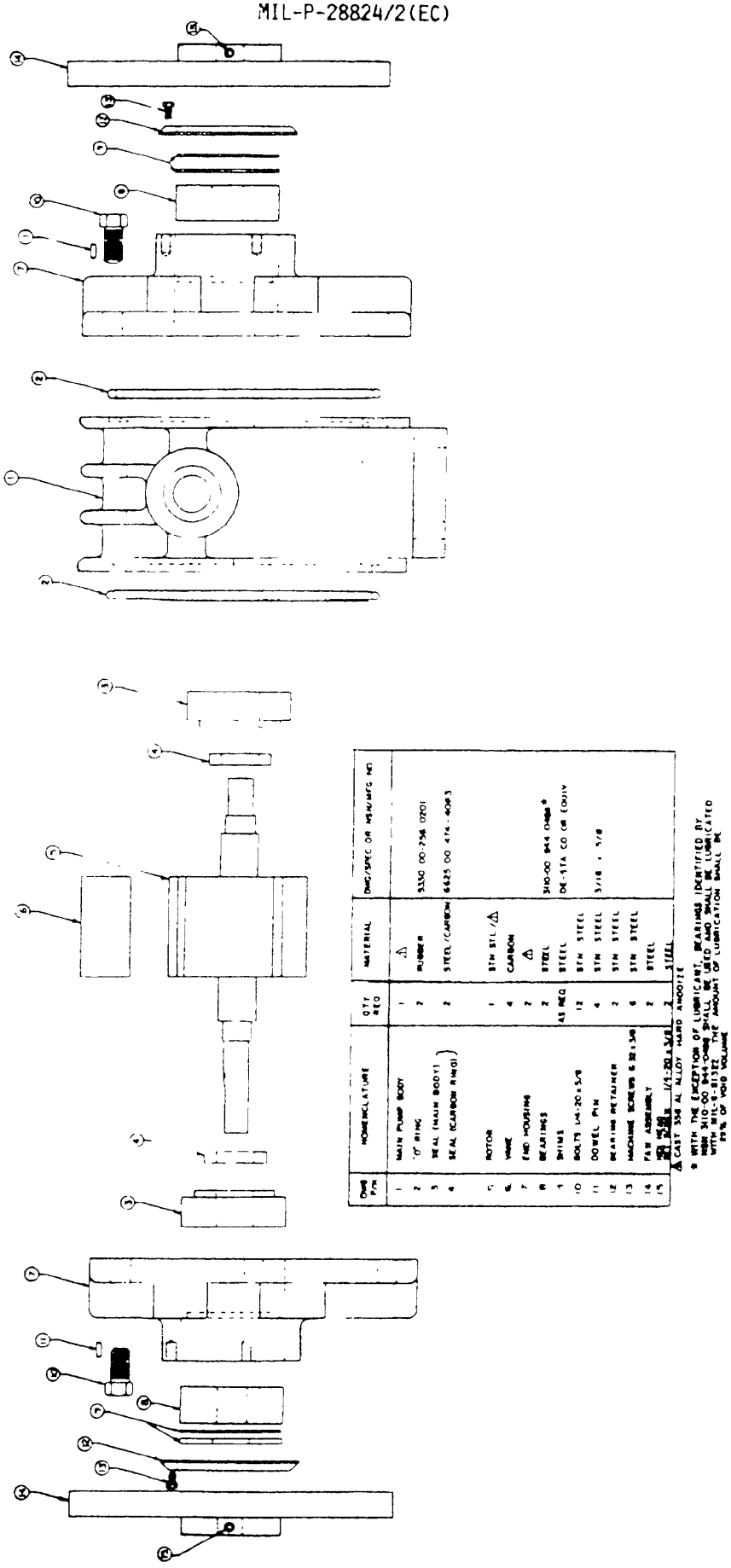


FIGURE 1 Pump assembly.

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Dwg Ref	NOMENCLATURE	QTY REQ	MATERIAL	DWG/SPEC OR MFR/MFG NO
1	MAIN PUMP BODY	1	⚠	5330 00-256 0201
2	'O' RING	2	RUBBER	
3	SEAL (MAIN BODY)	2	STEEL/CARBON	6625 00 474 - 4083
4	SEAL (CARBON RING)			
5	ROTOR	1	STN STL ⚠	
6	VANE	4	CARBON	
7	END HOUSING	2	⚠	
8	BEARINGS	2	BRONZE	3110-00 844 (MBB)*
9	SHIMS	AS REQ	STEEL	DE-STA CD CR (COAT)
10	BOLTS 1/4-20 x 5/8	12	STN STEEL	
11	DOWEL PIN	4	STN STEEL	5/16 x 5/8
12	BEARING RETAINER	2	STN STEEL	
13	MACHINE SCREWS 6-32 x 3/8	6	STN STEEL	
14	FAN ASSEMBLY	2	STEEL	
15	SEAL PLATE 1/4-20 x 3/4	2	STEEL	

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* WITH THE EXCEPTION OF LUBRICANT, BEARINGS IDENTIFIED BY MSBR 3110-00 844-(MBB) SHALL BE USED AND SHALL BE LUBRICATED WITH MIL-G-81322. THE AMOUNT OF LUBRICATION SHALL BE 75% OF VOID VOLUME

FIGURE 2. Pump exploded view.

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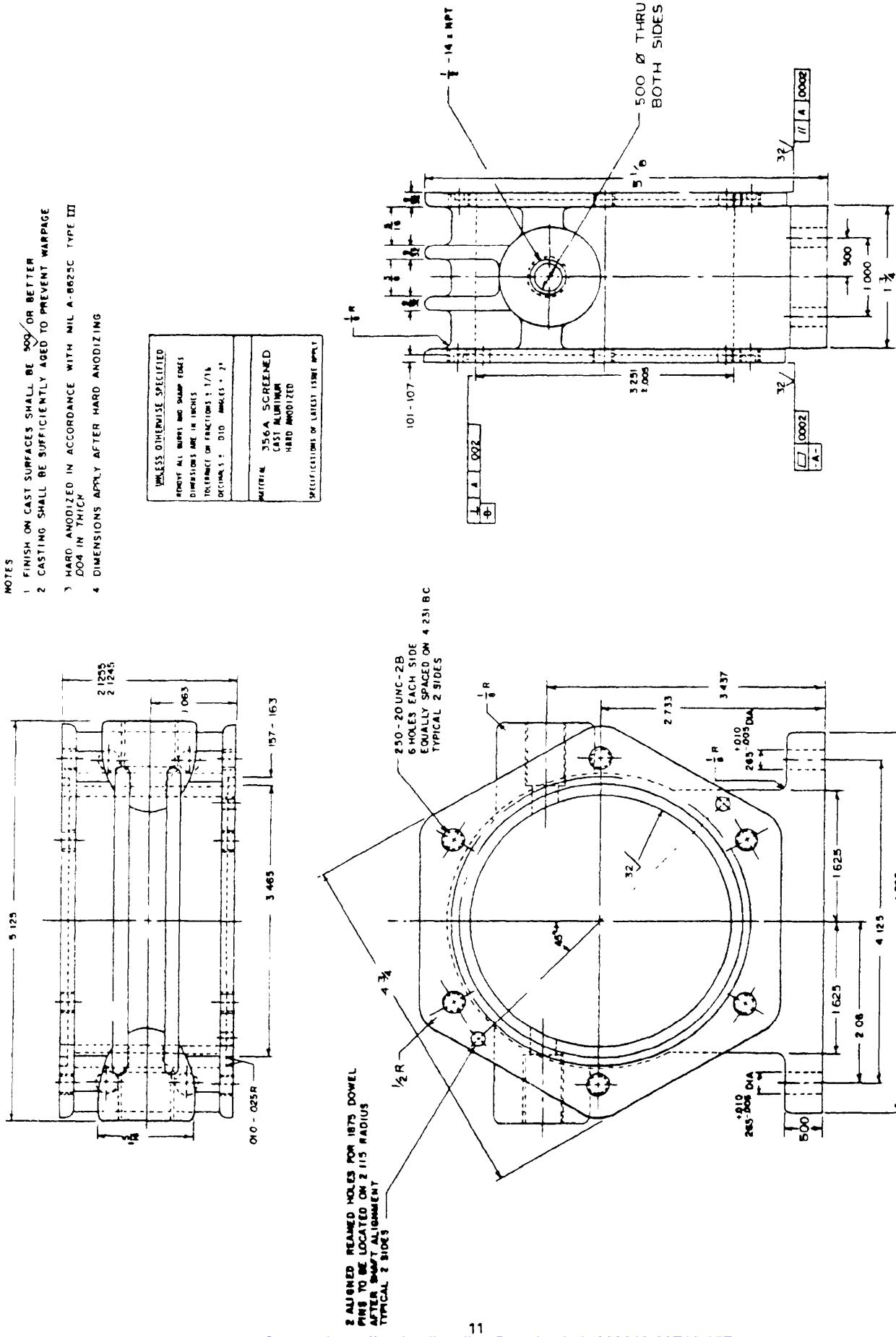
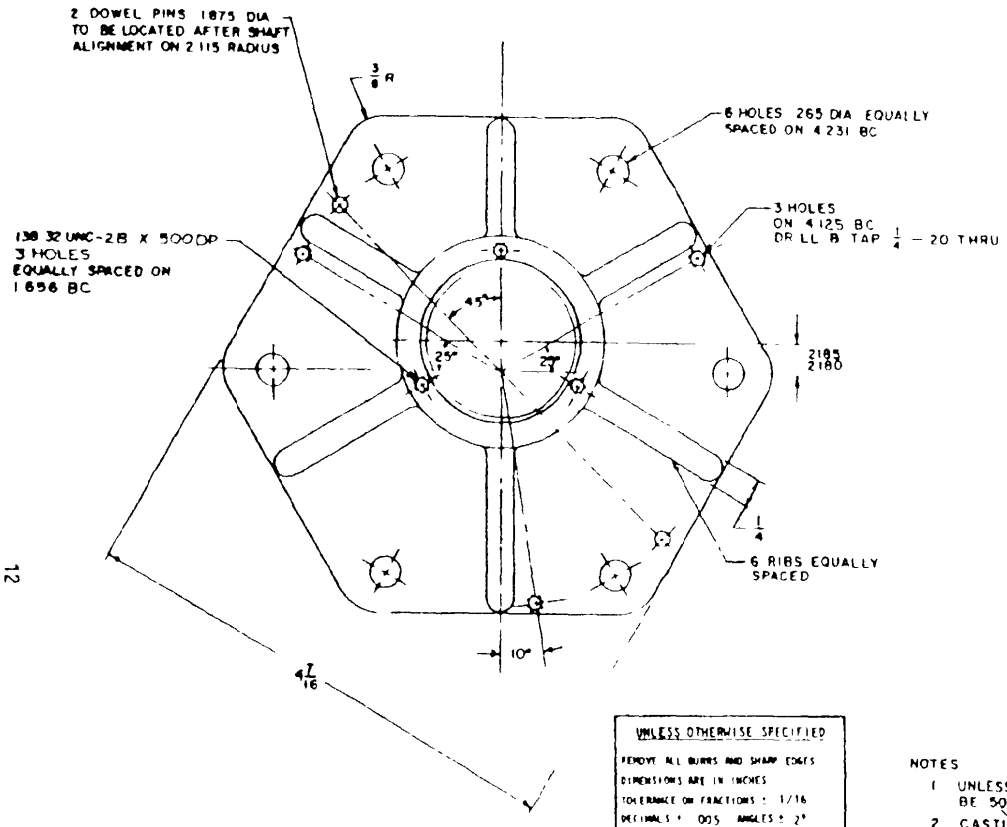
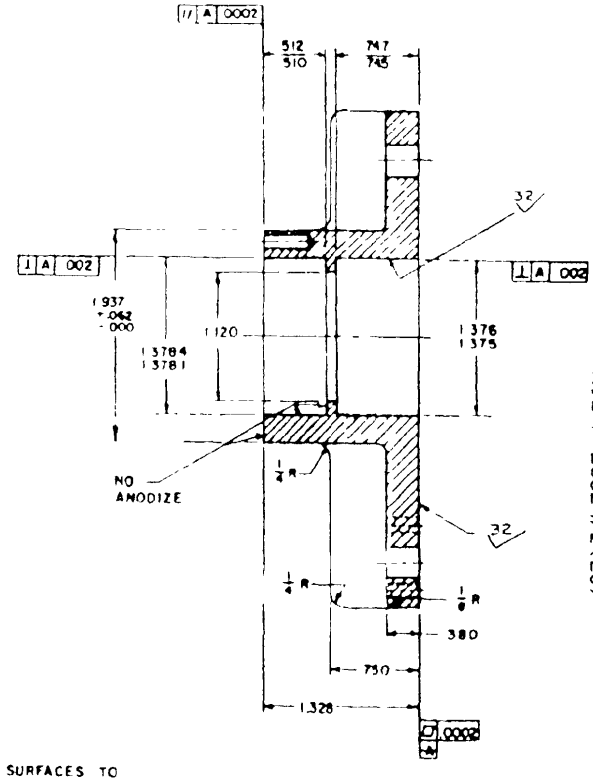


FIGURE 3. Pump main body.

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UNLESS OTHERWISE SPECIFIED
REMOVE ALL BURRS AND SHARP EDGES DIMENSIONS ARE IN INCHES TOLERANCE ON FRACTIONS ± 1/16 DECIMALS ± .005 ANGLES ± 2°
MATERIAL 356A SCREENED CAST ALUMINUM HARD ANODIZE
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NOTES

1. UNLESS OTHERWISE SPECIFIED ALL SURFACES TO BE 500 OR BETTER
2. CASTING SHALL BE SUFFICIENTLY AGED TO PREVENT WARPAGE
3. TWO (2) REQ'D PER PUMP ASSEMBLY
4. HARD ANODIZED IN ACCORDANCE WITH MIL-A-8625C TYPE III .004 INCH THICKNESS
5. DIMENSIONS APPLY AFTER HARD ANODIZING.

FIGURE 4 End housing

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NOTES:
1. UNLESS OTHERWISE SPECIFIED ALL SURFACES SHALL BE 16
2. FACES OF ROTOR SHALL BE PARALLEL WITHIN .0005
3. HARD ANODIZED IN ACCORDANCE WITH MIL-A-8625C TYPE III, .004 IN. THICK
4. SHAFT AND ROTOR ARE TWO SEPARATE PIECES. ROTOR MAY BE CAST ON SHAFT OR ATTACHED TO SHAFT IN ANY ACCEPTABLE MANNER SHAFT IS STAINLESS STEEL ROTOR IS ALUMINUM HARD ANODIZED AS PER NOTE 3.

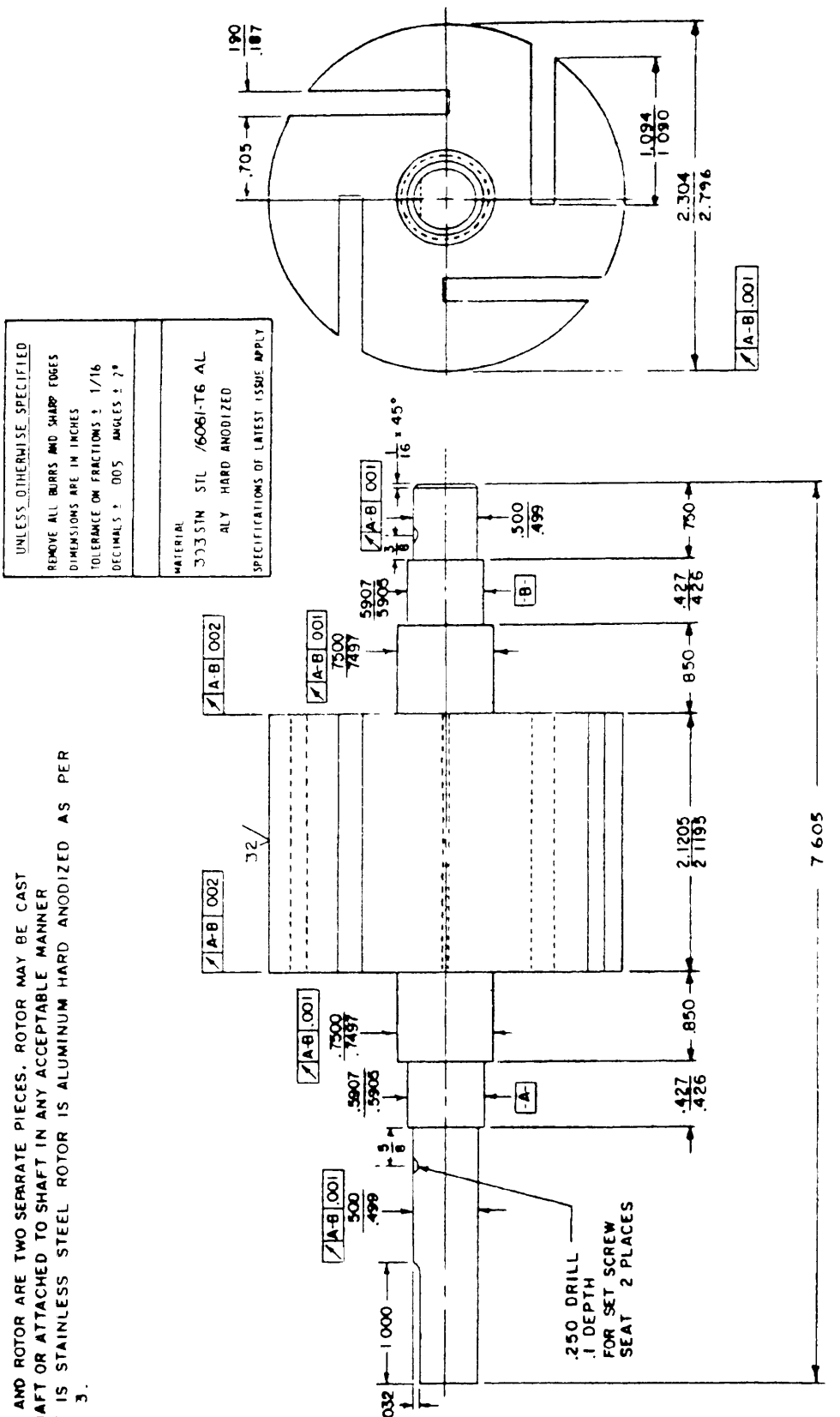


FIGURE 5. Rotor.

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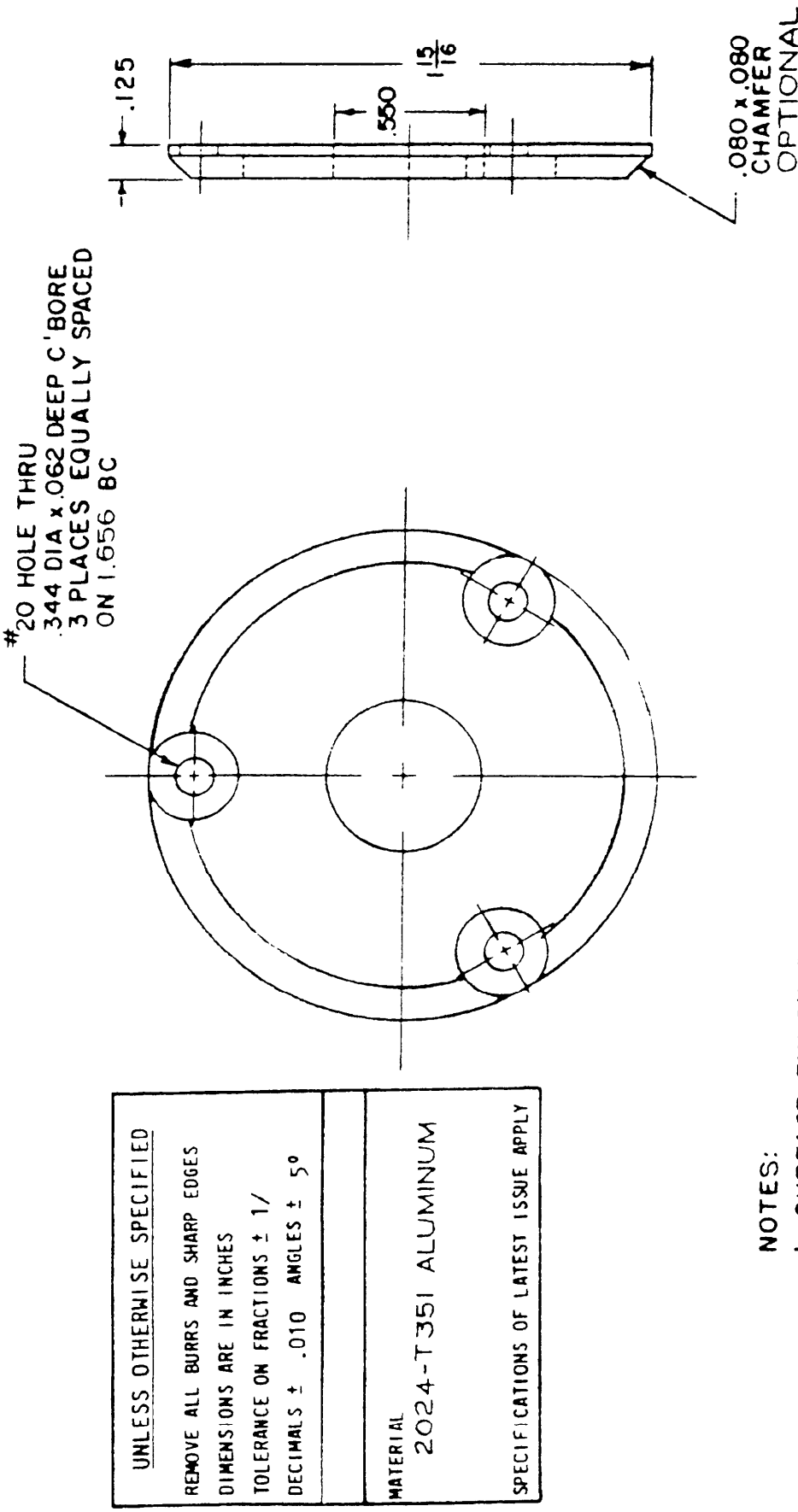
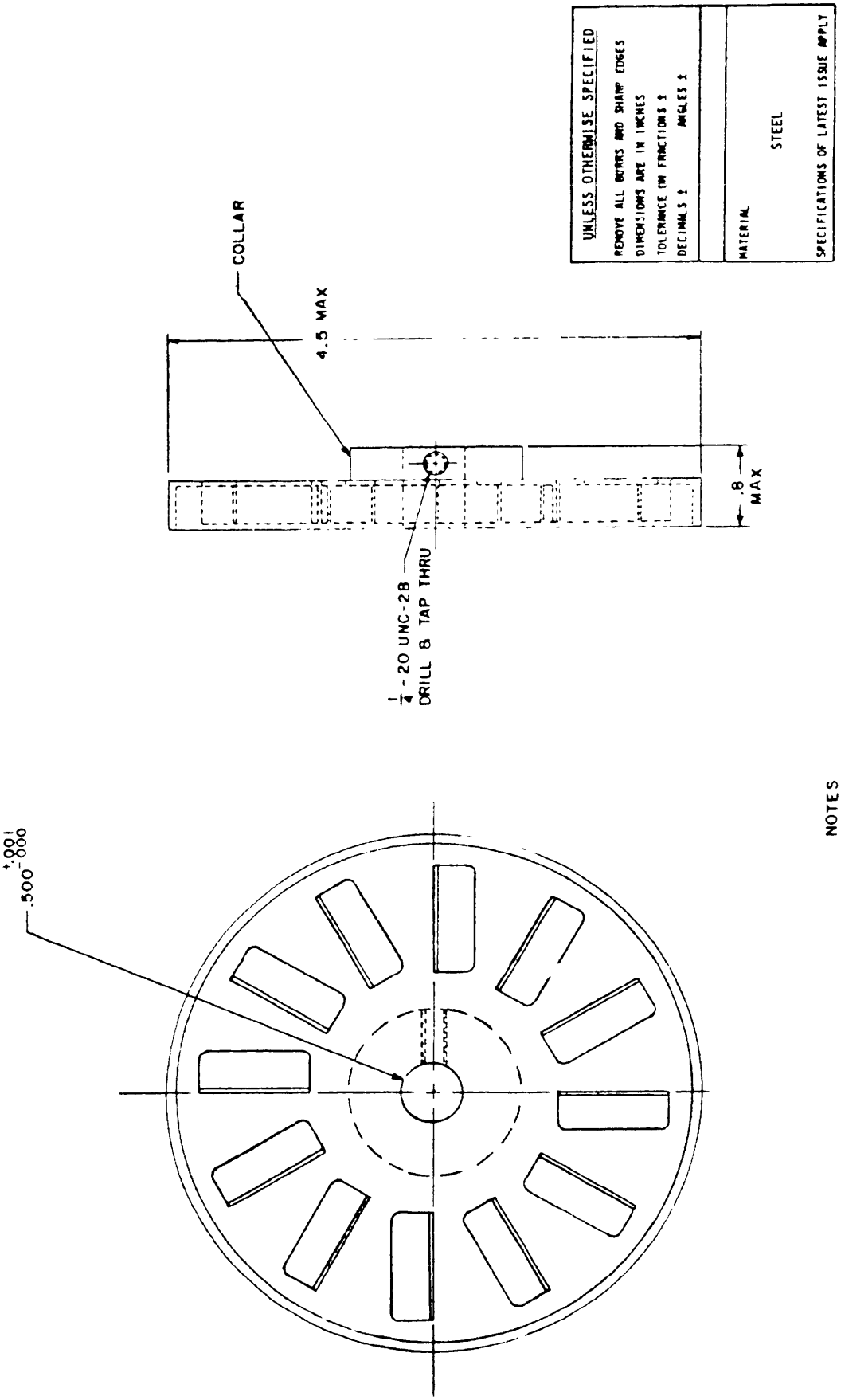


FIGURE 6. Bearing retainer.

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NOTES

1. FINISH SHALL BE 125 OR BETTER.
2. FAN DESIGN MAY BE OPTIONAL EXCEPT WHERE DIMENSIONS ARE GIVEN.

FIGURE 7. Fan.

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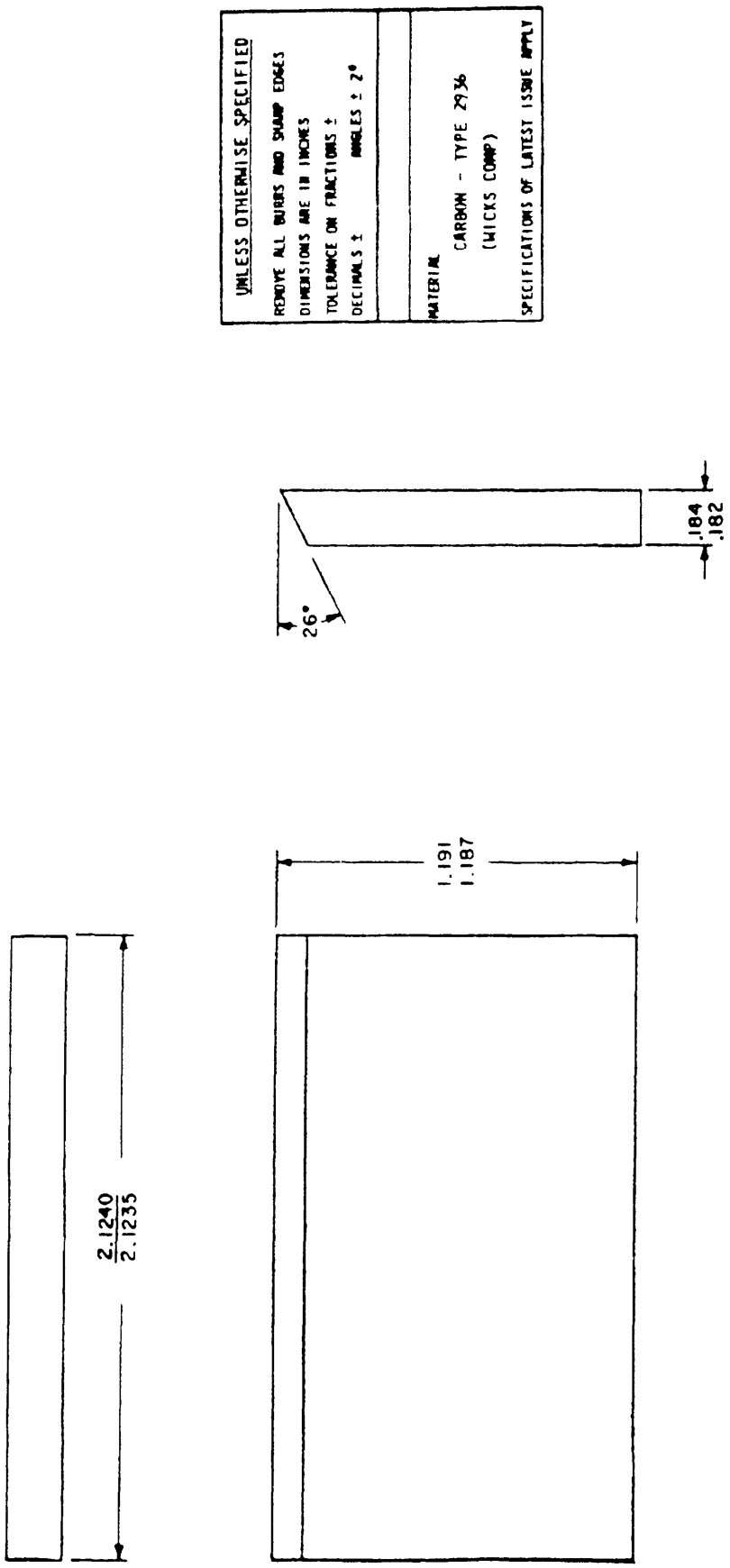


FIGURE 8. Vane.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-28824/_ (EC)	Pump, Vacuum, Positive Displacement, Air Rotary Vane, Type 2, Cast Aluminum

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify): _____
b. ADDRESS (Street, City, State, ZIP Code)	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
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